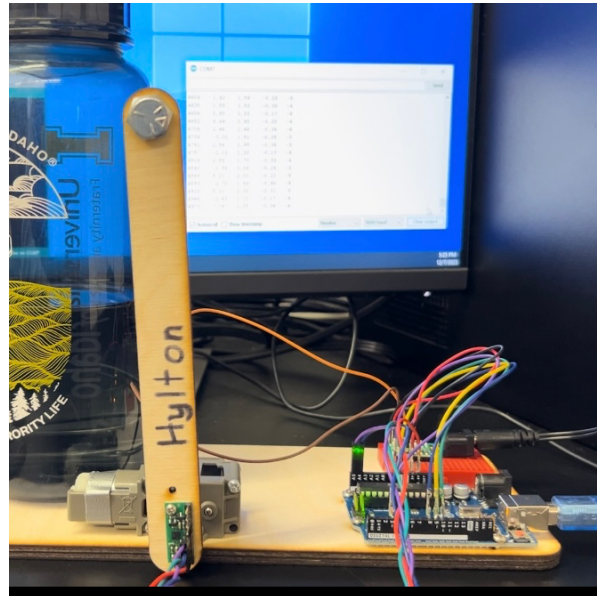


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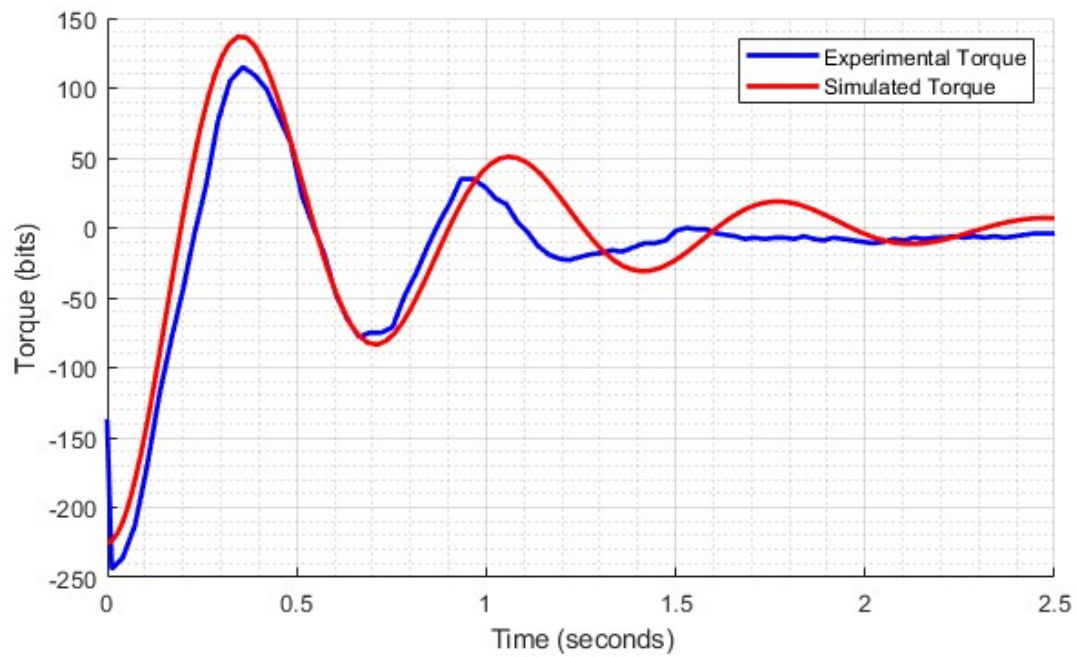
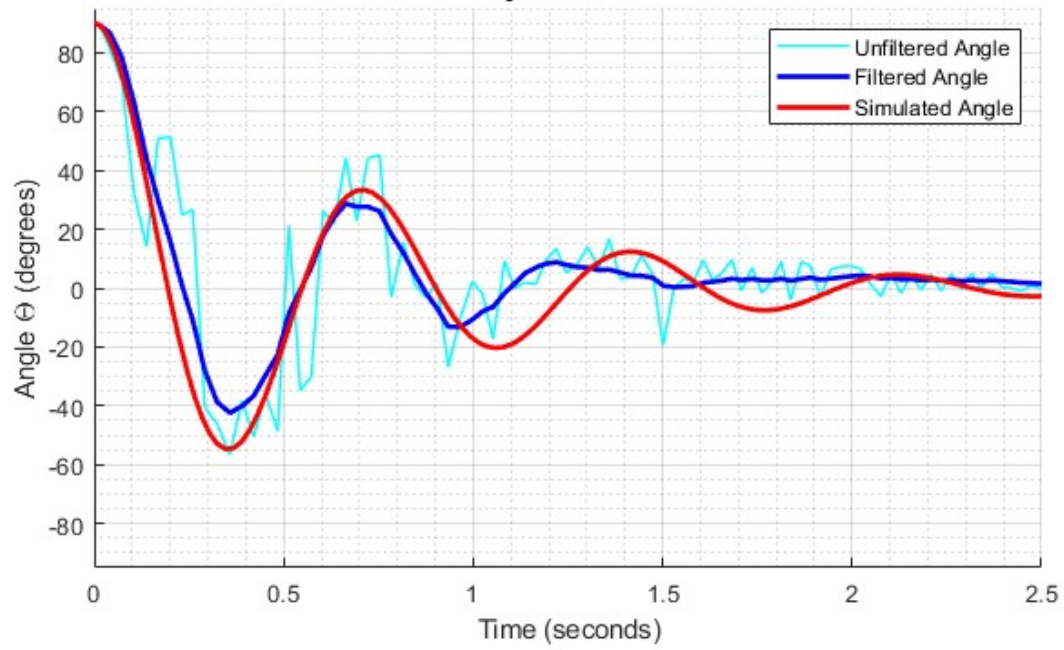
December 7, 2023

Final Pendulum Project



Variable	Value	Units
Complimentary filter, c	0.9	-
Static Gain for Overshoot, kd	2.725	Bits/Degrees
Derivative Gain for Overshoot, kp	0	Bits*Seconds/Degrees
Static Gain for Tuned, kd	4	Bits/Degrees
Derivative Gain for Tuned, kp	0.225	Bits*Seconds/Degrees

ME 313 Project #4 Overshoot Experiment Hylton Reber



I was not able to get my pendulum to print the data from the tuned experiment. It became too jittery and would not work unless I did not begin the Serial and didn't print the data.

The purpose of this project was to learn how the proportional and derivative gains affect the output of a transient system. To do this, I experimented with the values of the proportional and derivative gains by changing the values and observing the outcome. The largest issue came from the pendulum jittering around instead of having smooth motion, and my motor breaking several times and stripping the pendulum attachment which delayed me significantly.

This report describes results of the 50% overshoot experiment and simulated data results and methods, the effect of the complimentary filter, and the tuned experiment data results and methods.

To find the 50% overshoot, I started with a proportional gain value of 1 and increased it slowly until my pendulum began overshooting. I ran into some issues with the motor being too active and changing directions too quickly, which I overcame by trouble shooting my code and lowering the proportional gain value. An issue that quickly arose was the pendulum jittering around instead of going anywhere, I fixed this by basically trying a bunch of changes in my Arduino code until something worked. I found that a proportional gain value of 2.725 would give me, on average, the desired 50% overshoot. This was not something that would regularly occur, sometimes it would overshoot by almost 100% and other times it would overshoot by 10%, I have not come to any conclusions as to how to make the 50% overshoot occur more consistently. After getting the overshoot data, I analyzed the plots and used transient response equations to find the zeta and natural frequency values and adjusted them so the simulated data became closer to the experimental data, the values that worked were: $\zeta = 0.1$, $\omega_n = 14$, and a static gain = 0.565 (adjusted from the static gain found in project 3 experiment A).

To find the tuned value of proportional gain and derivative gain, I started with a proportional gain of 4 and a derivative gain of 1. From there, I slowly adjusted the derivative gain until the pendulum rose to zero degrees at a quick but smooth rate. I was not able to print the data without the system becoming too jittery and not working.

In conclusion, I've run out of time and should finish my future projects days before they are due, I am sorry to both you and my grade.